3.5 SINGLE-PHASE FULL-WAVE RECTIFIER WITH A HIGHLY INDUCTIVE LOAD

With a resistive load, the input current of the single-phase rectifier will be a sine wave. With an inductor load, the input current will be distorted as shown in Figure 3.4c. If the load is highly inductive, the load current will remain almost constant with a small amount of ripple content and the input current will be like a square wave. Let us consider the waveforms of Figure 3.8, where v_s is the sinusoidal input voltage, i_s is the instantaneous input current, and i_{s1} is its fundamental component.

If ϕ is the angle between the fundamental components of the input current and voltage, ϕ is called the *displacement angle*. The *displacement factor* is defined as

$$\text{DF} = \cos \phi \tag{3.21}$$

The *harmonic factor* (HF) of the input current is defined as

$$\text{HF} = \left(\frac{I_s^2 - I_{s1}^2}{I_{s1}^2}\right)^{1/2} = \left[\left(\frac{I_s}{I_{s1}}\right)^2 - 1\right]^{1/2} \tag{3.22}$$

where I_{s1} is the fundamental component of the input current I_s. Both I_{s1} and I_s are expressed here in rms. The input *power factor* (PF) is defined as

$$\text{PF} = \frac{V_s I_{s1}}{V_s I_s} \cos \phi = \frac{I_{s1}}{I_s} \cos \phi \tag{3.23}$$

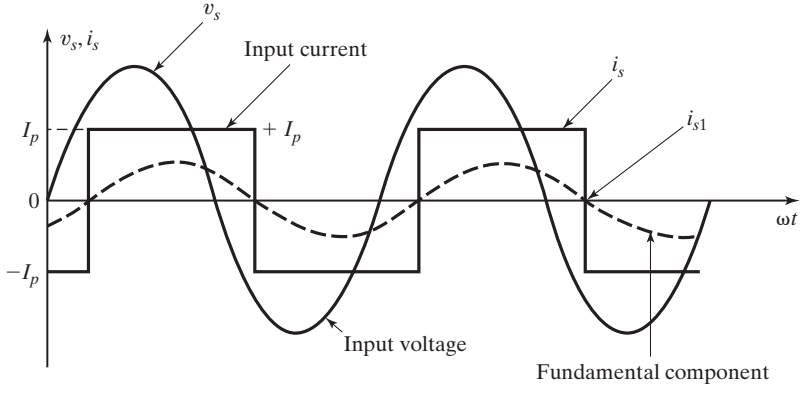


FIGURE 3.8

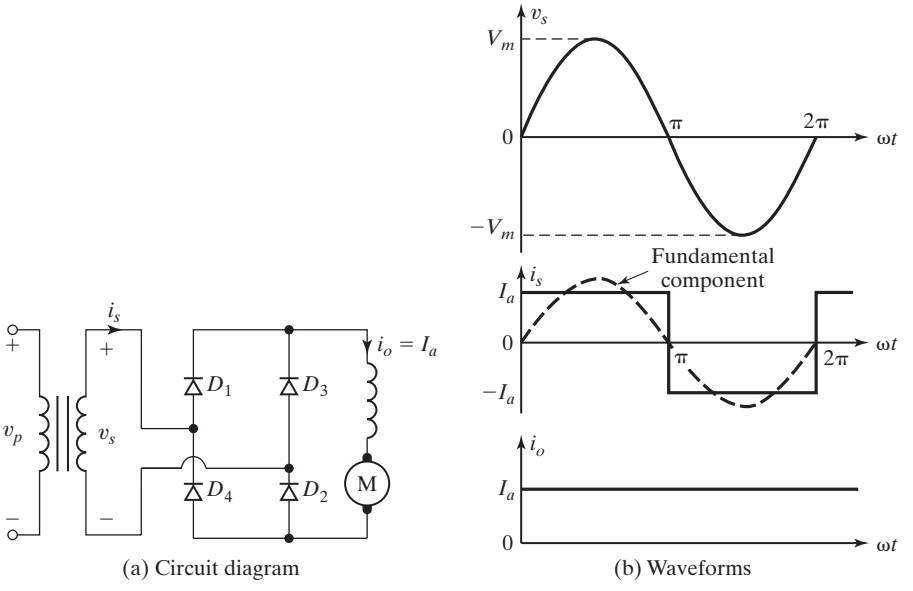
Waveforms for input voltage and current.

Notes

1. HF is a measure of the distortion of a waveform and is also known as *total harmonic distortion* (THD).
2. If the input current i_s is purely sinusoidal, $I_{s1} = I_s$ and the power factor PF equals the displacement factor DF. The displacement angle ϕ becomes the impedance angle $\theta = \tan^{-1}(\omega L/R)$ for an RL load.
3. Displacement factor DF is often known as *displacement power factor* (DPF).
4. An ideal rectifier should have $\eta = 100\%$, $V_{ac} = 0$, RF $= 0$, TUF $= 1$, HF $=$ THD $= 0$, and PF $=$ DPF $= 1$.

Example 3.4 Finding the Input Power Factor of a Full-Wave Rectifier

A single-phase bridge rectifier that supplies a very high inductive load such as a dc motor is shown in Figure 3.9a. The turns ratio of the transformer is unity. The load is such that the motor draws a ripple-free armature current of I_a as shown in Figure 3.9b. Determine (a) the HF of input current and (b) the input PF of the rectifier.



(a) Circuit diagram (b) Waveforms

FIGURE 3.9

Full-wave bridge rectifier with dc motor load.

Solution

Normally, a dc motor is highly inductive and acts like a filter in reducing the ripple current of the load.

a. The waveforms for the input current and input voltage of the rectifier are shown in Figure 3.9b. The input current can be expressed in a Fourier series as

$$i_s(t) = I_{dc} + \sum_{n=1,3,\ldots}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

where

$$I_{dc} = \frac{1}{2\pi}\int_0^{2\pi} i_s(t)\, d(\omega t) = \frac{1}{2\pi}\int_0^{2\pi} I_a\, d(\omega t) = 0$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} i_s(t) \cos n\omega t\, d(\omega t) = \frac{2}{\pi}\int_0^{\pi} I_a \cos n\omega t\, d(\omega t) = 0$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} i_s(t) \sin n\omega t\, d(\omega t) = \frac{2}{\pi}\int_0^{\pi} I_a \sin n\omega t\, d(\omega t) = \frac{4I_a}{n\pi}$$

Substituting the values of a_n and b_n, the expression for the input current is

$$i_s(t) = \frac{4I_a}{\pi}\left(\frac{\sin \omega t}{1} + \frac{\sin 3\omega t}{3} + \frac{\sin 5\omega t}{5} + \cdots\right) \tag{3.24}$$

The rms value of the fundamental component of input current is

$$I_{s1} = \frac{4I_a}{\pi\sqrt{2}} = 0.90 I_a$$

The rms value of the input current is

$$I_s = \frac{4}{\pi\sqrt{2}} I_a \left[1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{7}\right)^2 + \left(\frac{1}{9}\right)^2 + \cdots\right]^{1/2} = I_a$$

From Eq. (3.22),

$$\text{HF} = \text{THD} = \left[\left(\frac{1}{0.90}\right)^2 - 1\right]^{1/2} = 0.4843 \quad \text{or} \quad 48.43\%$$

b. The displacement angle $\phi = 0$ and $\text{DF} = \cos\phi = 1$. From Eq. (3.23), the $\text{PF} = (I_{s1}/I_s)\cos\phi = 0.90$ (lagging).

Key Points of Section 3.5

- The input power factor of a rectifier with a resistive load is PF = 1.0 and PF = 0.9 for a highly inductive load. The power factor will depend on the load inductor and the amount of distortion of the input current.